

Unifying Special Relativity and Quantum Mechanics

Lecture 24 — Spontaneous Symmetry Breaking

(The global $SO(2)$ scalar model, the Mexican hat, the vacuum manifold,
shifted fields, the massless Goldstone mode,
and the non-linear realization of the symmetry)

Typed version of the handwritten notes, with added explanations in blue boxes

Added explanation (not in the original notes) — How to read this document

Conventions. Exactly as in Lectures 1–23: normal text and the yellow / pink / grey boxes are a faithful transcription of the handwritten sheets, and the text colours follow the pens of the original (blue, magenta, green, red, pencil). Everything inside a blue box like this one is material I have added; it is not in the original notes.

Pagination. This scan opens a brand-new chapter, *Spontaneous symmetry breaking*, and the page numbering restarts at 1. Since the previous chapter (Lectures 22–23) carried eleven stars, these sheets are marked with *twelve*: [p. 1*****]–[p. 22*****]. **Good news for once:** the scan is complete, strictly in order, with no duplicated sheets, no missing sheets and no overlap with Lecture 23. The circled numbers 1–22 in the top-right corner of each sheet run consecutively, and the PDF page number equals the sheet number throughout. If you are following the paper scan alongside this document, sheet n is simply [p. n*****].

Three transcription slips flagged in place.

1. On [p. 1*****] the invariant is written $\phi^T \phi = \phi_1^2 + \phi_i^2$; the index on the second term should of course be 2 (Section 1).
2. On [p. 11*****] the first kinetic term is written $\frac{1}{2}(\partial_\mu \phi_1)(\partial^\mu \phi_2)$; it should be $\frac{1}{2}(\partial_\mu \phi_1)(\partial^\mu \phi_1)$, as the right-hand side of the same equation confirms (Section 11).
3. On [p. 12*****]–[p. 13*****] the field θ occasionally carries a stray exponent (a θ^4 where θ^2 is meant, and vice versa) — pure handwriting drift; the corrected powers are used here and noted at the point where they occur.

Two symbols that look alike. From [p. 15*****] onwards the notes use *two* different “rho”s: $\varrho(x)$ for the *complex* scalar field, and $\rho(x)$ for the *real radial* field in the exponential parametrization. In the handwriting they are nearly indistinguishable. I keep the distinction rigidly: ϱ is complex, ρ is real and positive. Watch for it, because $\varrho = \frac{1}{\sqrt{2}}\rho e^{i\pi/f_\pi}$ is the central formula of the second half of the lecture.

Notation for the shifted fields. The lecture does the same calculation *twice*, with two different names for the fields. In the Cartesian parametrization (Part III) the massive field is χ and the massless one is θ . In the exponential parametrization (Part IV) the massive field is h and the massless one is π . They are the same two particles; $\chi \leftrightarrow h$ and $\theta \leftrightarrow \pi$ agree to leading order but differ at the level of interactions. Section 14 explains why the lecture bothers to do it twice.

Added explanation (not in the original notes) — Where we left off — recap of Lecture 23

Lecture 23 finished the construction of non-abelian gauge theory and then broke it. It showed that for a *semi-simple* group $G = G_1 \times \cdots \times G_m$ the adjoint representation is reducible ($f^{ABC} = 0$ between different factors), so the Yang–Mills Lagrangian splits into one block per simple factor, each with its own independent coupling g_i ; with additional $U(1)$ factors one simply adds the ordinary abelian field strength. Matter was then coupled through tensor-product representations, and Fermi theory was recovered as the low-energy limit.

The lecture ended on a *failure*. Giving a Yang–Mills boson a mass by hand makes the longitudinal polarization physical; nothing then organizes its interactions, and amplitudes grow like $g^2 E^2/M^2$, so the theory loses unitarity at $E \sim M/g$. Something must appear below that scale. The lecture stopped there, deliberately, on the question “*what are these new states?*”

This lecture is the first half of the answer. The resolution is not to abandon the mass but to obtain it *without* writing a symmetry-breaking term in the Lagrangian: one keeps a perfectly symmetric Lagrangian and lets the *vacuum* fail to be symmetric. That is spontaneous symmetry breaking, and Lecture 24 develops it in its simplest possible setting — two real scalars with a global $SO(2)$ symmetry, no gauge fields anywhere. The gauge fields come back in the next lecture, when this mechanism is combined with local symmetry to give the Higgs mechanism. Do not expect to see a gauge boson in these 22 sheets; the point here is the appearance of a *massless scalar*, which is precisely the mode that will later be eaten.

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PART I — The classical setup: a scalar theory with global $SO(2)$ symmetry [p. 1 *****]–[p. 3 *****]

1 Spontaneous symmetry breaking: two fields packed into a vector [p. 1 *****]

■ SPONTANEOUS SYMMETRY BREAKING

We start from a classical discussion.

■ Spontaneous breaking of an $SO(2)$ global symmetry

We start from a scalar field theory with global $SO(2)$ symmetry.

We take two scalar fields, $\phi_1(x)$ and $\phi_2(x)$, and we pack them into a two-components object

$$\phi(x) = \begin{pmatrix} \phi_1(x) \\ \phi_2(x) \end{pmatrix}$$

that transforms according to the fundamental representation of $SO(2)$ (that is, it rotates like a vector)

$$\phi(x) = \begin{pmatrix} \phi_1(x) \\ \phi_2(x) \end{pmatrix} \xrightarrow{SO(2)} R(\theta) \begin{pmatrix} \phi_1(x) \\ \phi_2(x) \end{pmatrix}$$

$$\triangleright R(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

The quantity

$$\phi^T \phi = (\phi_1 \quad \phi_2) \begin{pmatrix} \phi_1 \\ \phi_2 \end{pmatrix} = \phi_1^2 + \phi_2^2$$

is clearly invariant under $SO(2)$, and it will serve as “building block” of the Lagrangian density.

Added explanation (not in the original notes)

Slip flagged. The handwritten line reads $\phi^T \phi = \phi_1^2 + \phi_i^2$; the subscript on the second term is a slip for 2. Corrected above.

Why $\phi^T \phi$ is the only building block. $R(\theta)$ is a real orthogonal matrix, $R^T R = \mathbb{1}$, so $\phi^T \phi \rightarrow (R\phi)^T (R\phi) = \phi^T R^T R \phi = \phi^T \phi$. Nothing else built from two undifferentiated ϕ ’s is invariant: the antisymmetric combination $\epsilon_{ij} \phi_i \phi_j$ vanishes identically. So $\phi^T \phi$ is the *unique* quadratic invariant, and every $SO(2)$ -invariant potential is a function of it alone. That single fact is what forces the potential to be a function of the *radius* in field space — which is exactly why the minimum can be a whole circle rather than a point.

What “global” means here, and why it matters later. θ is a single constant number, the same at every space-time point. Contrast this with the *local* symmetries of Lectures 15 and 22–23, where $\theta = \theta(x)$ and a gauge field had to be introduced. This lecture deliberately keeps the symmetry global: no gauge fields appear anywhere in the 22 sheets. The combination of spontaneous breaking with a *local* symmetry is the Higgs mechanism, and it is the subject of the next lecture. Keeping them separate is the right pedagogical order, because the massless field that appears here is precisely what gets eaten there.

2 The Lagrangian, and the search for the lowest-energy configuration [p. 2*****]

We consider the Lagrangian

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi)^T (\partial^\mu \phi) - \frac{m^2}{2} \phi^T \phi - \frac{\lambda}{4} (\phi^T \phi)^2 \quad \star$$

$\partial_\mu \phi = \begin{pmatrix} \partial_\mu \phi_1 \\ \partial_\mu \phi_2 \end{pmatrix}$; since the symmetry is global,

$$\partial_\mu \phi \longrightarrow \partial_\mu (R\phi) = R (\partial_\mu \phi),$$

and the kinetic term is invariant.

↪ Most general renormalizable Lagrangian invariant under $SO(2)$ global.

We now look for the field configuration of lowest energy.

▷ The discussion is classical, but very often we shall use a “quantum language”, and we shall say that we look for “the vacuum state of the theory”.

To find such field configuration, we look at the Hamiltonian of the theory.

The Hamiltonian associated to $\star$ is

Added explanation (not in the original notes)

Why the quartic is the last term allowed. In four space-time dimensions a scalar field has mass dimension 1, so $(\phi^T \phi)^n$ has dimension $2n$ and its coupling has dimension $4 - 2n$. For $n = 2$ the coupling λ is dimensionless — marginal, renormalizable. For $n = 3$ it would carry dimension -2 : non-renormalizable, excluded by the classification of Lecture 13. That is the whole content of the word “renormalizable” in the blue line above.

On the sign conventions. Note that m^2 is written as a single symbol, and the lecture will shortly allow it to be *negative*. This is deliberate and slightly abusive: m^2 is a parameter in the Lagrangian, not the square of anything. Only in Case 1 does it coincide with a physical mass-squared. In Case 4, where $m^2 = -\mu^2 < 0$, the physical masses are read off only *after* the fields have been shifted — and they come out positive. Writing a “negative mass squared” in the Lagrangian is not a sickness of the theory; it is a statement that you expanded around the wrong point.

3 The Hamiltonian and the potential; the four cases [p. 3*****]

$$H = \int d^3 \vec{x} \left[\sum_{k=1}^2 \frac{1}{2} (\partial_0 \phi_k) (\partial_0 \phi_k) + \frac{1}{2} \sum_{k=1}^2 \vec{\nabla} \phi_k \cdot \vec{\nabla} \phi_k + V \right]$$

$$V(\phi_1, \phi_2) = \frac{m^2}{2} (\phi_1^2 + \phi_2^2) + \frac{\lambda}{4} (\phi_1^2 + \phi_2^2)^2$$

↪ “potential” of the theory

The first two terms are always positive and are minimized to zero if we consider field configurations that are constant in space and time.

↪ Consequently, the key is the minimization of the potential energy.

We set $X \equiv (\phi_1^2 + \phi_2^2)^{1/2}$, so that

$$V(X) = \frac{m^2}{2}X^2 + \frac{\lambda}{4}X^4$$

$$V'(X) = m^2X + \lambda X^3 = X(m^2 + \lambda X^2) \stackrel{!}{=} 0 \longrightarrow \begin{cases} X = 0 \\ X^2 = -\frac{m^2}{\lambda} \end{cases}$$

$$V''(X) = m^2 + 3\lambda X^2$$

■ There are different cases depending on the values of the parameters. Let us discuss all of them.

Added explanation (not in the original notes)

What X is, and why the problem is one-dimensional. $X = |\phi|$ is the *radius* in the two-dimensional field space (ϕ_1, ϕ_2) . Because the potential depends only on the invariant $\phi^T \phi = X^2$, the full two-dimensional minimization collapses to a one-variable problem in $X \geq 0$. This is the whole trick of the next three pages. But keep in mind what has been thrown away: a single value of X corresponds to a whole *circle* of field configurations. When $X = 0$ that circle degenerates to a point; when $X \neq 0$ it does not. That is the entire origin of spontaneous symmetry breaking, and it is invisible in the $V(X)$ plots — you have to go back to field space (which the notes do on [p. 6*****]–[p. 8*****]) to see it.

Why constant configurations. The first two terms of H are $\frac{1}{2}\dot{\phi}_k^2$ and $\frac{1}{2}|\vec{\nabla}\phi_k|^2$, both manifestly non-negative. They vanish precisely when ϕ_k is independent of t and $\vec{x}$. Since we are minimizing a sum of non-negative terms plus V , we may set them to zero first and then minimize V over constants. This is why the vacuum problem is an ordinary calculus problem and not a variational one.

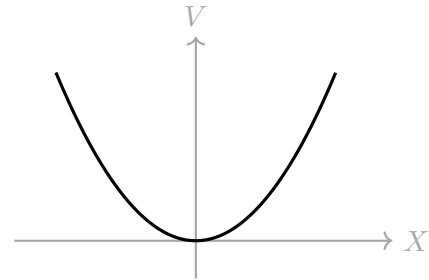
The four cases, in advance. With two parameters and two signs there are four combinations; $\lambda > 0$ is required for the potential to be bounded below at large X (the $\lambda X^4/4$ term dominates), which immediately kills Cases 2 and 3. Of the two survivors, $m^2 > 0$ gives a unique symmetric vacuum and $m^2 < 0$ gives the interesting one.

PART II — The four cases and the shape of the potential

[p. 4*****]–[p. 8*****]

4 Cases 1, 2 and 3 [p. 4*****]

Case 1: $m^2 > 0, \lambda > 0$



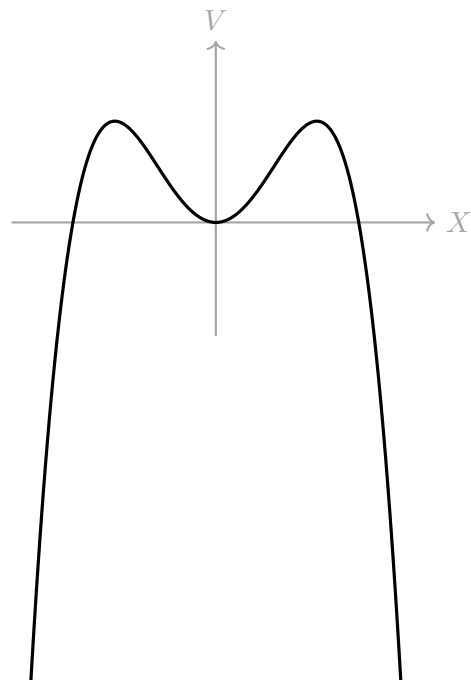
(redrawn from the hand sketch)

The potential has a minimum at $X = 0$, that is for

$$\phi_1 = \phi_2 = 0$$

The minimum is unique.

Case 2: $m^2 > 0, \lambda < 0$



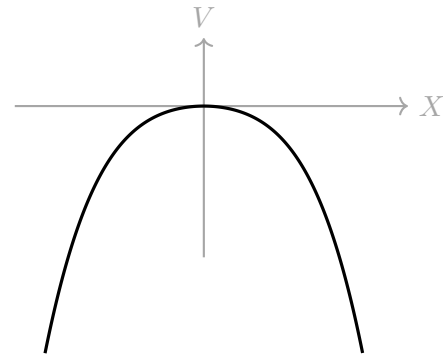
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Minimum at $X = 0$, but only a local one.

Maximum at $X^2 = -m^2/\lambda$, then it is unbounded from below at large field values.

↔ We discard this situation.

Case 3: $m^2 < 0, \lambda < 0$



(redrawn from the hand sketch)

We also discard this situation, for the same reason.

Added explanation (not in the original notes)

Reading the three sketches. All that distinguishes them is the sign of λ and of m^2 , and the decisive one is λ :

Case 1 ($m^2 > 0, \lambda > 0$): both terms positive, $V \geq 0$ with the single stationary point $X = 0$. $V''(0) = m^2 > 0$: a genuine minimum. The second root $X^2 = -m^2/\lambda$ is negative, hence unphysical — there is nothing else.

Case 2 ($m^2 > 0, \lambda < 0$): $X = 0$ is still a local minimum, but the second root $X^2 = -m^2/\lambda$ is now positive and is a maximum, beyond which $\lambda X^4/4 \rightarrow -\infty$. The state at the origin is metastable at best; the theory has no ground state at all.

Case 3 ($m^2 < 0, \lambda < 0$): both terms push down. $X = 0$ is a maximum and there is nothing below.

Why “unbounded from below” is fatal. A Hamiltonian with no lower bound has no ground state: one can extract arbitrarily much energy by running the field to infinity, and there is nothing to expand around, so the quantum theory cannot even be defined perturbatively. The requirement $\lambda > 0$ is therefore not an aesthetic choice — it is a consistency condition on the theory. (Strictly, $\lambda > 0$ is required only for the potential; $\lambda = 0$ gives a free theory.)

5 Case 4: the interesting one [p. 5*****]

Case 4: $m^2 \equiv -\mu^2, \mu^2 > 0; \lambda > 0$

The stationary point with $X = 0$ has

$$V''(X=0) = m^2 = -\mu^2 < 0 \rightsquigarrow \underline{\text{maximum at } X=0}$$

■ The stationary point with $X^2 = -\frac{m^2}{\lambda} = +\frac{\mu^2}{\lambda}$ has

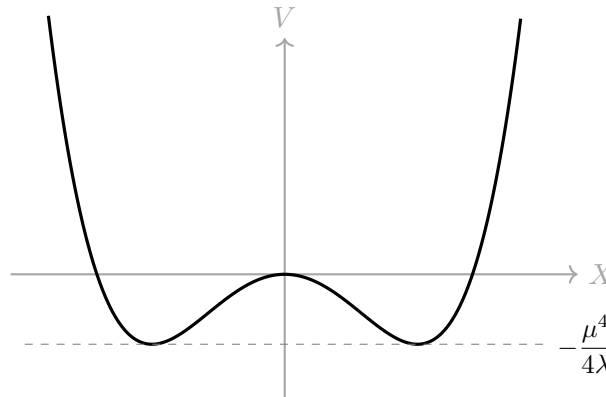
$$V''\left(X^2 = \frac{\mu^2}{\lambda}\right) = m^2 + 3\lambda\left(-\frac{m^2}{\lambda}\right) = m^2 - 3m^2 = -2m^2 = +2\mu^2 > 0$$

$$\rightsquigarrow \underline{\text{minimum at } X^2 = \frac{\mu^2}{\lambda}}$$

■ The value of the potential at the minima is

$$V\left(X^2 = \frac{\mu^2}{\lambda}\right) = \frac{m^2}{2} \left(-\frac{m^2}{\lambda}\right) + \frac{\lambda}{4} \frac{m^4}{\lambda^2} = -\frac{m^4}{2\lambda} + \frac{m^4}{4\lambda} = -\frac{m^4}{4\lambda} = -\frac{\mu^4}{4\lambda}$$

The situation, as function of “ X ”, is



In order to have a vacuum state with zero energy, we add to the potential a constant shift $+\mu^4/4\lambda$.

The situation, therefore, is

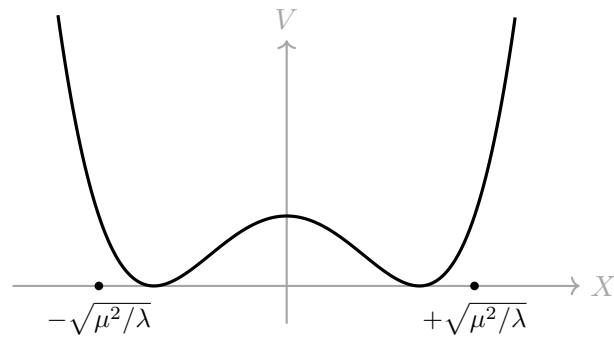
Added explanation (not in the original notes)

Two remarks on the arithmetic. (i) Note the trick in V'' : since $\lambda X^2 = -m^2$ at the second stationary point, $3\lambda X^2 = -3m^2$, so $V'' = m^2 - 3m^2 = -2m^2$. In Case 4 that is $+2\mu^2 > 0$. Keep this number: it is *exactly* the mass-squared of the physical field that will emerge on [p. 13*****] and [p. 19*****], $m_\chi^2 = m_h^2 = 2\mu^2$. That is no coincidence — the mass-squared of a fluctuation is by definition the second derivative of the potential at the minimum. (ii) The two stationary points have swapped roles compared with Case 1: the origin, which used to be the minimum, is now a maximum.

Why one is allowed to add a constant. A constant in $\mathcal{L}$ does not change any equation of motion, any Feynman rule, or any S -matrix element — $\delta S/\delta\phi$ is blind to it. In a theory without gravity it is pure bookkeeping, chosen here so that the vacuum energy is zero and the potential at its minimum touches the axis. *With* gravity it is not bookkeeping at all: a constant energy density is a cosmological constant, and it gravitates. The fact that this constant is freely adjustable here, but observed to be fantastically small in nature, is the cosmological constant problem. Nothing in this lecture depends on the choice.

6 The shifted potential, and Case 1 in field space [p. 6*****]

$$V(X) = -\frac{\mu^2}{2} X^2 + \frac{\lambda}{4} X^4 + \frac{\mu^4}{4\lambda}$$



Let us focus on [Case 1](#) and [Case 4](#), in field space.

$$m^2 > 0, \lambda > 0 \quad (\text{Case 1})$$

Minimum at $\phi_1 = \phi_2 = 0$.

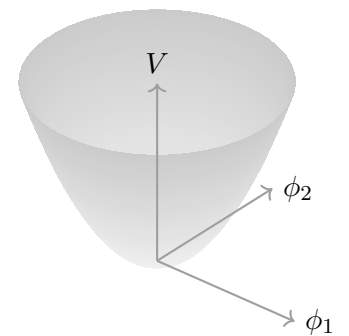
The vacuum is unique and, most importantly, invariant under the $SO(2)$ global symmetry.

↔ This is a trivial statement. The vacuum is

$$\phi_0 = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

and if we apply a rotation we get

$$R(\theta) \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$



(redrawn from the hand sketch)

This is precisely the situation in which the symmetry is realized à la Wigner–Weyl.

7 Case 1 in the spectrum; Case 4 and the manifold of vacua

[p. 7*****]

In this case, the Lagrangian is manifestly $SO(2)$ invariant, and the symmetry is manifest in the spectrum of the theory.

The Lagrangian

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi)^T \partial^\mu \phi - \frac{m^2}{2} \phi^T \phi - \frac{\lambda}{4} (\phi^T \phi)^2$$

describes two scalar particles with the same mass m (mass degeneracy in the spectrum is a trademark of a symmetry realized à la Wigner–Weyl)

$$m^2 = -\mu^2 < 0; \lambda > 0 \quad (\text{Case 4})$$

The situation is drastically different.

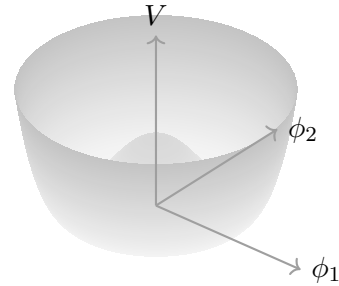
The vacuum is not unique.

This is because in field space the condition

$$\phi_1^2 + \phi_2^2 = \frac{\mu^2}{\lambda}$$

identifies a manifold of vacua.

The minimum occurs for any ϕ_1 and ϕ_2 such that $\phi_1^2 + \phi_2^2 = \mu^2/\lambda$, that is for any ϕ_1 and ϕ_2 that lie on a circle of radius $(\mu^2/\lambda)^{1/2}$.



(redrawn from the hand sketch — the “Mexican hat”)

Added explanation (not in the original notes)

Where the mass degeneracy of Case 1 comes from. Expand V around $\phi = 0$: the quadratic part is $\frac{m^2}{2}(\phi_1^2 + \phi_2^2)$, so both fields get the *same* m^2 . This is forced: the only quadratic invariant is $\phi^T \phi$, and it weights ϕ_1 and ϕ_2 equally. The notes return to this on [p. 14*****].

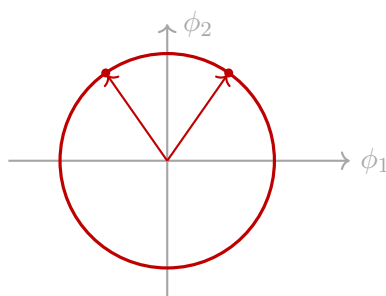
Read this against Lecture 12. There, “symmetry realized à la Wigner–Weyl” meant: G is represented by unitary operators $U(g)$ on the Hilbert space, the vacuum satisfies $U(g)|0\rangle = |0\rangle$, and one-particle states fall into multiplets that are irreducible representations of G — necessarily degenerate in mass, since $[Q^A, P^\mu] = 0$. The two scalars here are the $SO(2)$ doublet. Everything that follows is a study of what happens when the very first of those conditions, $U(g)|0\rangle = |0\rangle$, fails.

The geometry of the Mexican hat. V depends only on $X = |\phi|$, so the surface is a *surface of revolution*: take the double well of [p. 6*****] and spin it about the V axis. The maximum at $X = 0$ becomes the central bump; the two minima at $X = \pm\sqrt{\mu^2/\lambda}$ sweep out one circular trough. This is why the vacuum is a one-parameter family: the angular direction costs nothing.

8 The vacuum manifold S^1 , and the spontaneous choice [p. 8*****]

vacuum manifold S^1

$$\phi_1^2 + \phi_2^2 = \mu^2/\lambda$$



(redrawn from the hand sketch)

All points on the circle are equivalent, and related by $SO(2)$ rotations.

What happens in this situation?

We have a degeneracy:

a continuum of ground states,
all with the same energy, and
all connected via $SO(2)$ rotations.

The system spontaneously chooses one of the degenerate vacua.

Suppose that you keep a ball at the maximum of the potential, and let it go. The ball will fall towards the bottom of the potential, and will eventually come to rest at some specific point on the circle $\phi_1^2 + \phi_2^2 = \mu^2/\lambda$.

Which point we choose on the circle to be the ground state of the theory is arbitrary (they are all equivalent!).

Added explanation (not in the original notes)

“ S^1 ” is the notation for a circle — the one-sphere, the set of points at fixed distance from the origin in a plane. The vacuum manifold is the set of field configurations that minimize V ; here it is a copy of S^1 of radius $v = \sqrt{\mu^2/\lambda}$.

The group-theoretic statement. $SO(2)$ acts *transitively* on this circle: any vacuum can be reached from any other by a rotation, and no rotation (except the identity) leaves a given point fixed. In the general language, the vacuum manifold is the coset G/H where H is the subgroup that leaves the chosen vacuum invariant. Here $G = SO(2)$, $H = \{\mathbb{I}\}$, and $G/H = SO(2) \simeq S^1$. The symmetry is *completely* broken. Counting: $\dim G - \dim H = 1 - 0 = 1$, and there will be exactly 1 massless field. Hold on to that number — it reappears on [p. 13*****] and [p. 19*****].

Why “spontaneous”. Nothing in the Lagrangian prefers one point on the circle. The asymmetry is not put in by hand; it arises because the system must *be* somewhere, and being somewhere on a circle is not being rotationally invariant. The familiar analogue is a ferromagnet: the microscopic Hamiltonian is rotationally invariant, but below the Curie temperature the magnetization points in some direction. Also a thin vertical rod compressed from above: the equations are axially symmetric, the buckled rod is not.

A caveat about the ball. The picture is classical and slightly misleading. In the quantum theory one might expect tunnelling between the degenerate vacua, as happens in ordinary quantum mechanics with a double well — in which case the true ground state would be a symmetric superposition and nothing would break. It does not happen here because these are *field* configurations: to tunnel from one vacuum to another one must change the field over *all of space*, and the tunnelling amplitude goes like $e^{-\text{const} \times V_{\text{space}}}$, which vanishes in infinite volume. This is why spontaneous symmetry breaking is a genuine phenomenon in field theory (infinitely many degrees of freedom) and not in ordinary quantum mechanics. The different vacua sit in orthogonal, mutually inaccessible Hilbert spaces — *superselection sectors* — and one simply picks one and works in it.

PART III — Consequences: shifted fields and the massless mode

[p. 9*****]–[p. 14*****]

9 Making a choice: the vacuum expectation value [p. 9*****]

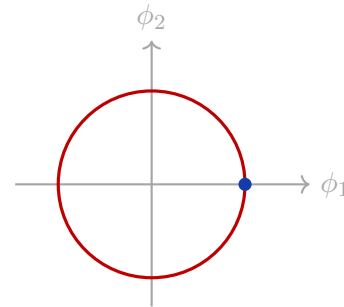
However, once a choice is made, the ground state loses its rotational invariance.

↪ We call this effect spontaneous breakdown of the symmetry.

Let's make a choice. We consider the point

$$\phi_1 = \sqrt{\frac{\mu^2}{\lambda}}, \quad \phi_2 = 0$$

We introduce the notation



$$v^2 \equiv \frac{\mu^2}{\lambda} \quad \langle \phi \rangle = \begin{pmatrix} v \\ 0 \end{pmatrix}$$

$$\langle \phi_1 \rangle = v, \quad \langle \phi_2 \rangle = 0$$

↪ We say that “the field ϕ has vacuum expectation value (VEV, for short) equal to $\begin{pmatrix} v \\ 0 \end{pmatrix}$ ”.

We use the word “VEV” even though this is not the expectation value of an operator on the ground state, but just the classical configuration describing the ground state. We will also use the notation

$$\langle \phi \rangle = \begin{pmatrix} v \\ 0 \end{pmatrix} \equiv \phi_0$$

Added explanation (not in the original notes)

Why this particular point, and why it costs nothing. Any point on the circle would do; choosing the one on the positive ϕ_1 axis is a choice of coordinates in field space, exactly like choosing where to put the z axis when you quantize angular momentum. All physical results must be — and are — independent of it. Concretely, if you chose the point at angle α instead, you would define $\phi' = R(-\alpha)\phi$ and repeat everything verbatim.

The pedantry in the pencil paragraph is worth taking seriously. In the classical discussion $\langle \phi \rangle$ is just the value of the field at the minimum. In the quantum theory it really *is* $\langle 0 | \hat{\phi} | 0 \rangle$, computed in the chosen vacuum $|0\rangle$, and the classical value is its leading approximation. The notes will exploit exactly this on [p. 10*****]–[p. 11*****]: the reason for shifting the fields is a statement about $\langle 0 | \hat{\phi} | 0 \rangle$ as an operator equation.

Order of magnitude, for orientation. In the Standard Model the analogous quantity is $v \simeq 246$ GeV, fixed by the measured Fermi constant. Everything about the weak scale is that one number.

10 The vacuum is not invariant; the shifted fields [p. 10*****]

We now see explicitly that the choice of the vacuum is not invariant under the symmetry.

If we apply a rotation

$$\phi_0 \longrightarrow \phi'_0 = R(\theta)\phi_0 = R(\theta) \begin{pmatrix} v \\ 0 \end{pmatrix} \neq \begin{pmatrix} v \\ 0 \end{pmatrix}$$

It seems we are in a situation conceptually different compared to the case in which the symmetry is realized à la Wigner–Weyl.

↔ In particular, the vacuum is not invariant under the action of the symmetry.

■ Consequences of spontaneous symmetry breaking

We define the so-called “shifted fields”

$$\begin{aligned}\phi_1(x) &\equiv v + \chi(x) \\ \phi_2(x) &\equiv \theta(x)\end{aligned}$$

with, by construction

$$\langle \chi(x) \rangle = \langle \theta(x) \rangle = 0$$

This is an important step for the definition of the theory at the quantum level.

↔ The key aspect is the following. When we consider the fields as operators, and define

11 Why the vev must be shifted away: creation operators and Fock space [p. 11*****]

annihilation and creation operators via $a(\vec{p})|0\rangle = 0$, $\langle 0|a^\dagger(\vec{p}) = 0$, we also have $\langle 0|\phi(x)|0\rangle = 0$, since

$$\phi(x) = \int \frac{d^3\vec{p}}{(2\pi)^3 \sqrt{2E_p}} \left[a(\vec{p}) e^{-ip \cdot x} + a^\dagger(\vec{p}) e^{ip \cdot x} \right]$$

Creation and annihilation operators, in turn, are used to build multi-particle states and the Fock space of the quantum theory.

↔ If we consider a field with $\langle 0|\phi(x)|0\rangle \neq 0$, we would necessarily have $a(\vec{p})|0\rangle \neq 0$. This makes inconvenient to work with fields with non-zero VEV.

Kinetic terms

Since “ v ” is just a constant, it does not affect the kinetic terms. We thus have

$$\frac{1}{2}(\partial_\mu \phi_1)(\partial^\mu \phi_1) + \frac{1}{2}(\partial_\mu \phi_2)(\partial^\mu \phi_2) = \frac{1}{2}(\partial_\mu \chi)(\partial^\mu \chi) + \frac{1}{2}(\partial_\mu \theta)(\partial^\mu \theta)$$

Potential

Let us now compute the potential $V(\phi)$.

Added explanation (not in the original notes)

Slip flagged. The handwritten left-hand side of the yellow box reads $\frac{1}{2}(\partial_\mu \phi_1)(\partial^\mu \phi_2)$ for the first term. It must be $(\partial_\mu \phi_1)(\partial^\mu \phi_1)$ — the right-hand side, with its two separate squares, confirms it. Corrected above.

The argument in the magenta arrow, spelled out. The whole apparatus of perturbative QFT — Fock space, particle interpretation, Feynman rules — rests on a field whose free part is a sum of a and $a^\dagger$ acting on a vacuum annihilated by a . If $\langle 0|\hat{\phi}|0\rangle = v \neq 0$, then $\hat{\phi}$ contains a c -number piece and $\hat{\phi}|0\rangle$ has a component along $|0\rangle$ itself, so $a(\vec{p})|0\rangle \neq 0$ and $|0\rangle$ is not the Fock vacuum of $\hat{\phi}$. Nothing is *wrong* — it is just that the mode expansion, and with it every

propagator and Feynman rule, has to be rewritten. The cure is trivial: define $\chi = \phi_1 - v$, which does have zero vev, and quantize *that*. The particles of the theory are the excitations of χ and θ , not of ϕ_1 and ϕ_2 .

The same statement in one line. A field with a vev is being expanded about the wrong point. Shifting is nothing but Taylor-expanding the potential around the true minimum instead of around a maximum. Everything on the next three sheets is that Taylor expansion.

Note the notation clash. The symbol θ was used on [p. 1 *****] for the $SO(2)$ rotation *angle*; from [p. 10 *****] onwards it is the name of the second shifted *field*. The notes do not comment on this. The rotation angle does not appear again after [p. 10 *****], so no ambiguity survives, but it is worth noticing when reading the original.

12 The potential in terms of the shifted fields [p. 12 *****]— [p. 13 *****]

$$\begin{aligned} V(\phi_1, \phi_2) &= -\frac{\mu^2}{2}(\phi_1^2 + \phi_2^2) + \frac{\lambda}{4}(\phi_1^2 + \phi_2^2)^2 + \frac{\mu^4}{4\lambda} = \\ &= -\frac{\mu^2}{2}[(v + \chi)^2 + \theta^2] + \frac{\lambda}{4}[(v + \chi)^2 + \theta^2]^2 + \frac{\mu^4}{4\lambda} = \\ &= -\frac{\mu^2}{2}(v^2 + \chi^2 + 2v\chi + \theta^2) + \frac{\lambda}{4}(v^2 + \chi^2 + 2v\chi + \theta^2)^2 + \frac{\mu^4}{4\lambda} \end{aligned}$$

Let us organize these terms as follows.

■ Terms that do not depend on the fields

$$-\frac{\mu^2 v^2}{2} + \frac{\lambda v^4}{4} + \frac{\mu^4}{4\lambda} = -\frac{\mu^4}{2\lambda} + \frac{\lambda \mu^4}{4\lambda^2} + \frac{\mu^4}{4\lambda} = 0$$

↑ since $v^2 = \mu^2/\lambda$

■ Terms linear in the fields

$$-\frac{\mu^2}{2}(2v\chi) + \frac{\lambda}{4}(4v^3\chi) = -\mu^2 v\chi + \lambda v \left(\frac{\mu^2}{\lambda}\right) \chi = -\mu^2 v\chi + v\mu^2 \chi = 0$$

■ Quadratic terms

$$\begin{aligned} &-\frac{\mu^2}{2}(\chi^2 + \theta^2) + \frac{\lambda}{4}(2v^2\chi^2 + 2v^2\theta^2 + 4v^2\chi^2) = \\ &= -\frac{\mu^2}{2}(\chi^2 + \theta^2) + \frac{\lambda}{4}(6v^2\chi^2 + 2v^2\theta^2) = \\ &= \chi^2 \left(-\frac{\mu^2}{2} + \frac{3\lambda v^2}{2}\right) + \theta^2 \left(-\frac{\mu^2}{2} + \frac{\lambda v^2}{2}\right) = \\ &= \chi^2 \left(-\frac{\mu^2}{2} + \frac{3\lambda \mu^2}{2\lambda}\right) + \theta^2 \left(-\frac{\mu^2}{2} + \frac{\lambda \mu^2}{2\lambda}\right) = \\ &= \chi^2 \left(-\frac{\mu^2}{2} + \frac{3\mu^2}{2}\right) + \theta^2 \underbrace{\left(-\frac{\mu^2}{2} + \frac{\mu^2}{2}\right)}_0 = \frac{1}{2}\chi^2(2\mu^2) \end{aligned}$$

■ Cubic and quartic terms

$$\frac{\lambda}{4} \left(\chi^4 + \theta^4 + 2\chi^2\theta^2 + 4v\chi^3 + 4v\chi\theta^2 \right)$$

Added explanation (not in the original notes)

Slip flagged. In the handwriting the exponents on θ drift in two places on [p. 12*****]–[p. 13*****] (a θ^4 appears where θ^2 is meant, and once the other way round). The powers used above are the correct ones, and can be checked against the binomial expansion below.

The expansion, in full. Write $A \equiv v^2 + 2v\chi + \chi^2 + \theta^2$. Then A^2 contains, sorted by total field degree:

$$\underbrace{v^4}_0 + \underbrace{4v^3\chi}_1 + \underbrace{4v^2\chi^2 + 2v^2\chi^2 + 2v^2\theta^2}_2 + \underbrace{4v\chi^3 + 4v\chi\theta^2}_3 + \underbrace{\chi^4 + 2\chi^2\theta^2 + \theta^4}_4.$$

The χ^2 coefficient is $6v^2$ because χ^2 arises twice: from $2 \cdot v^2 \cdot \chi^2$ and from $(2v\chi)^2 = 4v^2\chi^2$. The θ^2 coefficient is only $2v^2$, from $2 \cdot v^2 \cdot \theta^2$ — there is no (something) $\times \theta$ cross-term, because θ enters A only quadratically. **That asymmetry is the whole story of this lecture:** it is why χ ends up massive and θ massless.

Why the constant and linear terms had to vanish. Neither cancellation is an accident. The constant vanishes because the shift $+\mu^4/4\lambda$ was chosen on [p. 5*****] precisely to make $V = 0$ at the minimum. The linear term vanishes because we are expanding about a *stationary point*: $\partial V/\partial\phi_i = 0$ there by construction. If a linear term had survived, it would have meant the expansion point was not the minimum — and, at the quantum level, that $\langle\chi\rangle \neq 0$ after all, contradicting the definition of the shifted field. Checking that the linear (“tadpole”) term cancels is the standard consistency test after any shift.

Reading off the masses. A term $\frac{1}{2}m^2\varphi^2$ in V (equivalently $-\frac{1}{2}m^2\varphi^2$ in $\mathcal{L}$) means a particle of mass m . Here:

$$V \supset \frac{1}{2}(2\mu^2)\chi^2 + \frac{1}{2} \cdot 0 \cdot \theta^2 \implies m_\chi^2 = 2\mu^2, \quad m_\theta^2 = 0.$$

And note $m_\chi^2 = 2\mu^2 = V''$ at the minimum, exactly the number computed back on [p. 5*****].

13 The broken-phase Lagrangian: one massive, one massless

[p. 13*****]

We thus find the following Lagrangian written in terms of the shifted fields (this is also called Lagrangian written in the “broken phase”)

$$\mathcal{L} = \underbrace{\frac{1}{2}(\partial_\mu\chi)(\partial^\mu\chi) - \frac{1}{2}(2\mu^2)\chi^2 + \frac{1}{2}(\partial_\mu\theta)(\partial^\mu\theta)}_{\text{free Lagrangian}} - \underbrace{\frac{\lambda}{4}(\chi^4 + \theta^4 + 4v\chi^3 + 2\chi^2\theta^2 + 4v\chi\theta^2)}_{\text{interactions}}$$

Consider the quadratic Lagrangian. We observe that:

The field χ has mass $m_\chi^2 = 2\mu^2$.
The field θ is massless.

↪ This is a crucial difference compared to

Added explanation (not in the original notes)

Count the degrees of freedom. Before: two real scalars, both of mass m . After: two real scalars, one of mass $\sqrt{2}\mu$ and one massless. The number of fields is unchanged — shifting is just a change of variables — but the *spectrum* has changed completely.

The geometric reading (this is the picture to keep). Sit at the chosen vacuum, the point $(v, 0)$ at the bottom of the trough of the Mexican hat, and ask what it costs to move.

- Moving *radially* — the χ direction — climbs the wall of the trough. It costs energy quadratically, with curvature $V'' = 2\mu^2$. Hence a massive particle.
- Moving *along the trough* — the θ direction — goes around the circle of degenerate minima. It costs *nothing*, because every point of the circle has the same energy. Zero curvature, hence a massless particle.

So the massless mode is not a coincidence of the algebra: it exists because the vacuum manifold is a continuum, and the flat directions of the potential are exactly the directions along that manifold.

This is Goldstone's theorem (Nambu, Goldstone, 1961–62), in its simplest instance: the spontaneous breaking of a continuous global symmetry produces one massless scalar — a *Goldstone boson* — for every broken generator. Here $\dim G - \dim H = 1 - 0 = 1$, and there is exactly one massless field, θ . The notes are about to derive the symmetry reason for it ([p. 14*****]–[p. 21*****]), which is the more illuminating route.

A structural warning about the interaction terms. Note that θ appears in the interactions *without* derivatives: θ^4 , $\chi^2\theta^2$, $v\chi\theta^2$. In the exponential parametrization of Part IV the very same theory will be written with π appearing *only* through $\partial_\mu\pi$. Both are correct; the non-derivative couplings here conspire so that physical amplitudes vanish at zero momentum. The exponential variables make that cancellation manifest, which is the reason the lecture does the whole calculation a second time.

14 Wigner–Weyl versus spontaneous breaking — and the open question [p. 14*****]

the case of Wigner–Weyl: the symmetry is not realized in the mass spectrum of the theory.

Wigner–Weyl realization:

doublet of fields ϕ_1 and ϕ_2 with the same mass.

⇓

Symmetry explanation: the mass term $m^2\phi^T\phi$ respects the $SO(2)$ symmetry only if the masses are degenerate ($m_1^2\phi_1^2 + m_2^2\phi_2^2$ is not $SO(2)$ -invariant!).

Spontaneous symmetry breaking:

one massive + one massless field.

⇓

Is there an explanation in terms of symmetry? Why do we have a massless field?

→ **Why do we have a massless field?**

To answer this question, we make a different choice of field variables.

Added explanation (not in the original notes)

Why the answer must be a symmetry argument. A vanishing mass is not stable under quantum corrections unless something protects it. If $m_\theta^2 = 0$ were merely an accident of the tree-level algebra, loops would generate $m_\theta^2 \neq 0$ and the massless particle would disappear. The fact that it survives to all orders means there is a symmetry forbidding the term θ^2 — and Part IV identifies it: the shift symmetry $\pi \rightarrow \pi + c$, which is the original $SO(2)$ acting in a disguised, non-linear way.

Where this is going, in one paragraph. The lecture is about to rewrite the same theory in *polar* coordinates in field space: $\varrho = \frac{1}{\sqrt{2}}\rho e^{i\pi/f_\pi}$ instead of $\varrho = \frac{1}{\sqrt{2}}(\phi_1 + i\phi_2)$. Because the potential depends only on $|\varrho| = \rho/\sqrt{2}$, the angular field π *cannot appear in the potential at all* — not just at quadratic order, but to all orders. Its masslessness then becomes obvious rather than accidental, and the residual symmetry (constant shifts of the angle) is visible on the page. The same reason, expressed geometrically: π is a coordinate along the vacuum manifold, and the potential is constant along the vacuum manifold.

PART IV — The exponential parametrization and the non-linear realization [p. 15*****]–[p. 22*****]

15 The complex-field formulation and the global $U(1)$ [p. 15*****]

We consider the Lagrangian

$$\mathcal{L} = (\partial_\mu \varrho^*)(\partial^\mu \varrho) - V(\varrho)$$

$$V(\varrho) = -\mu^2 \varrho^* \varrho + \lambda (\varrho^* \varrho)^2 + \frac{\mu^4}{4\lambda}$$

($\varrho(x)$ is now a complex scalar field)

$\hookrightarrow$ This theory is equivalent to our original Lagrangian if we set

$$\varrho(x) = \frac{1}{\sqrt{2}} (\phi_1(x) + i \phi_2(x))$$

$$\text{with } \varrho^* \varrho = \frac{1}{2} (\phi_1^2 + \phi_2^2)$$

$\hookrightarrow$ This Lagrangian has a global $U(1)$ symmetry under which

$$\varrho(x) \longrightarrow e^{-i\alpha} \varrho(x)$$

This is the same $SO(2)$ symmetry of the previous Lagrangian (remember that $SO(2) \cong U(1)$).

The symmetry is spontaneously broken (assuming $\mu^2 > 0$, $\lambda > 0$) since the potential admits degenerate vacua described by the condition

$$\varrho^* \varrho = \frac{\mu^2}{2\lambda}$$

Added explanation (not in the original notes)

Check that the two Lagrangians really are the same theory. With $\varrho = \frac{1}{\sqrt{2}}(\phi_1 + i\phi_2)$:

$$(\partial_\mu \varrho^*)(\partial^\mu \varrho) = \frac{1}{2}(\partial_\mu \phi_1)(\partial^\mu \phi_1) + \frac{1}{2}(\partial_\mu \phi_2)(\partial^\mu \phi_2), \quad \varrho^* \varrho = \frac{1}{2}(\phi_1^2 + \phi_2^2).$$

The kinetic terms match term by term; note that a *complex* field is canonically normalized with coefficient 1, not $\frac{1}{2}$, precisely because it packages two real fields. For the potential:

$$-\mu^2 \varrho^* \varrho = -\frac{\mu^2}{2}(\phi_1^2 + \phi_2^2), \quad \lambda (\varrho^* \varrho)^2 = \frac{\lambda}{4}(\phi_1^2 + \phi_2^2)^2,$$

exactly the potential of [p. 6*****]. And the minimum condition $\varrho^* \varrho = \mu^2/2\lambda$ is $\frac{1}{2}(\phi_1^2 + \phi_2^2) = \mu^2/2\lambda$, i.e. $\phi_1^2 + \phi_2^2 = v^2$ — the same circle. So $|\langle \varrho \rangle| = v/\sqrt{2}$.

Why $SO(2) \cong U(1)$. A rotation by θ in the (ϕ_1, ϕ_2) plane is

$$\phi_1 + i\phi_2 \longrightarrow (\cos \theta \phi_1 - \sin \theta \phi_2) + i(\sin \theta \phi_1 + \cos \theta \phi_2) = e^{i\theta}(\phi_1 + i\phi_2),$$

which is multiplication of ϱ by a phase. Both groups are the circle; the sign of α in the notes is a convention. This identification was already used in Lecture 12, where $U(1)$ charge and antiparticles were discussed, and in Lecture 15 for QED — what is new here is not the group but the fate of its vacuum.

Why bother with complex notation at all? Because the invariant $\varrho^* \varrho$ is manifestly the squared *modulus*, so the split of ϱ into modulus and phase — the next page — is natural. In Cartesian variables the same split would look like an unmotivated change of coordinates.

16 Radial and angular fields: the exponential parametrization

[p. 16*****]

The physics of SSB becomes more transparent if we perform a further field redefinition. We write

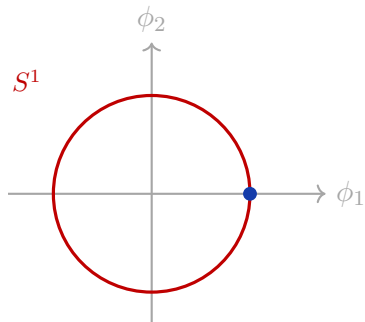
$$\varrho(x) = \frac{1}{\sqrt{2}} \left[\underbrace{\phi_1(x)}_{\text{radial field}} + i \underbrace{\phi_2(x)}_{\text{angular field}} \right] = \frac{1}{\sqrt{2}} \underbrace{\rho(x)}_{\text{radial field}} e^{i \underbrace{\pi(x)/f_\pi}_{\text{angular field}}}$$

“Cartesian parametrization”
in field space

“exponential parametrization”
in field space

“ f_π ” at this stage is a constant with mass-dimension 1, so that the field $\pi(x)$ also has mass-dimension 1.

We now make a choice of the vacuum that breaks the symmetry.



$$\begin{aligned} \langle \phi_1 \rangle &= v \\ \langle \phi_2 \rangle &= 0 \end{aligned}$$

in the exp. parametrization $\longrightarrow$

$$\begin{cases} \langle \rho(x) \rangle = v \\ \langle \pi(x) \rangle = 0 \end{cases}$$

Added explanation (not in the original notes)

Careful: two rhos. ϱ (complex) and ρ (real, positive) are different objects. In the handwriting they are the same squiggle. The relation is $|\varrho| = \rho/\sqrt{2}$, so $\langle \rho \rangle = v$ corresponds to $|\langle \varrho \rangle| = v/\sqrt{2}$, consistent with $\varrho^* \varrho = \mu^2/2\lambda = v^2/2$ from [p. 15*****].

These are polar coordinates in field space. $\rho \geq 0$ is the distance from the origin in the (ϕ_1, ϕ_2) plane, and π/f_π is the polar angle. Explicitly $\rho = \sqrt{\phi_1^2 + \phi_2^2} = X$, the very variable introduced on [p. 3*****]; and $\pi/f_\pi = \arctan(\phi_2/\phi_1)$. The change of variables is singular at the origin (as polar coordinates always are), which is harmless here because we expand around $\rho = v \neq 0$. It is also periodic: $\pi \rightarrow \pi + 2\pi f_\pi$ is the identity map, a global subtlety that does not affect perturbation theory.

Why f_π is needed. The exponent must be dimensionless. In four dimensions a scalar field has mass dimension 1, so a constant of mass dimension 1 must sit in the denominator if we

want $\pi(x)$ itself to be a canonically-normalizable field. On [p. 19*****] the requirement that π have a canonical kinetic term will fix $f_\pi = v$, so f_π is not a new parameter — it is being carried along as a symbol so that its role is visible.

Where the name comes from. f_π is the pion decay constant of QCD ($\simeq 92$ MeV), and the pions are the Goldstone bosons of the spontaneously broken chiral symmetry of the strong interactions. The letters π and f_π in these notes are a deliberate nod to that, the historically first and still the most important physical realization of this mechanism. Nothing in this lecture depends on the naming.

The key structural fact, before any algebra. V depends on ϱ only through $\varrho^* \varrho = \rho^2/2$. The phase cancels identically: $e^{-i\pi/f_\pi} e^{+i\pi/f_\pi} = 1$. Therefore $\pi(x)$ *cannot appear in the potential at all*. Its masslessness — indeed the absence of any non-derivative self-coupling — is now a one-line observation instead of a cancellation between terms.

17 Shifted fields; the potential [p. 17*****]–[p. 18*****]

We introduce shifted fields. We define

$$\rho(x) \equiv v + h(x), \quad \langle h(x) \rangle = 0$$

so that

$$\varrho(x) = \frac{1}{\sqrt{2}} [v + h(x)] e^{i\pi(x)/f_\pi}$$

We now write the Lagrangian in the broken phase, in terms of $h(x)$ and $\pi(x)$.

Potential

We start from the potential.

↔ We observe that

$$\varrho^* \varrho = \frac{1}{2} (v + h)^2$$

Since $V = -\mu^2 \varrho^* \varrho + \lambda (\varrho^* \varrho)^2 + \frac{\mu^4}{4\lambda}$, the field $\pi(x)$ completely disappears from the potential.

We write

$$\begin{aligned} V &= -\frac{\mu^2}{2} (v + h)^2 + \frac{\lambda}{4} (v + h)^4 = -\frac{\mu^2}{2} (v^2 + h^2 + 2vh) + \frac{\lambda}{4} (v^2 + h^2 + 2vh)^2 = \\ &= -\frac{\mu^2}{2} (v^2 + h^2 + 2vh) + \frac{\lambda}{4} (v^4 + h^4 + 4v^2 h^2 + 2v^2 h^2 + 4v^3 h + 4vh^3) + \frac{\mu^4}{4\lambda} = \\ &= \underbrace{h(-\mu^2 v + \lambda v^3)}_{=0} + h^2 \underbrace{\left(-\frac{\mu^2}{2} + \frac{3v^2 \lambda}{2}\right)}_{-\frac{\mu^2}{2} + \frac{3\lambda}{2} \left(\frac{\mu^2}{\lambda}\right) = \mu^2 = \frac{1}{2}(2\mu^2)} + \frac{\lambda}{4} h^4 + v\lambda h^3 \end{aligned}$$

$$V(h) = \frac{1}{2}(2\mu^2)h^2 + \frac{\lambda}{4}h^4 + (v\lambda)h^3$$

Added explanation (not in the original notes)

The struck-out terms. The three cancelled pieces are the field-independent ones: $-\frac{\mu^2}{2}v^2 + \frac{\lambda}{4}v^4 + \frac{\mu^4}{4\lambda} = 0$ with $v^2 = \mu^2/\lambda$, exactly the computation already done in Cartesian variables on [p. 12*****]. The strikes are in the original.

The two cancellations, again. The linear term is $-\mu^2vh + \lambda v^3h = vh(\lambda v^2 - \mu^2) = 0$, since $\lambda v^2 = \mu^2$: the tadpole vanishes, confirming that $\rho = v$ is the stationary point. The quadratic coefficient gives $m_h^2 = 2\mu^2$ — the same mass as χ in Part III, as it must be, since these are two coordinate systems for one theory.

And the point of the whole exercise: V depends on h alone. There is no π -anything — no π^2 , no π^4 , no $h\pi^2$. Compare the Cartesian result on [p. 13*****], where θ^4 , $\chi^2\theta^2$ and $v\chi\theta^2$ all appeared. The difference is not physics; it is that the exponential variables have moved every π interaction into the *kinetic* term, where it necessarily comes with derivatives. That is computed next.

18 Kinetic terms in the exponential parametrization [p. 18*****]— [p. 19*****]

Kinetic terms

We compute $(\partial_\mu \varrho^*)(\partial^\mu \varrho)$.

$$\begin{aligned} (\partial_\mu \varrho^*)\partial^\mu \varrho &= \frac{1}{2} \partial_\mu \left((h+v)e^{-i\pi/f_\pi} \right) \partial^\mu \left((h+v)e^{i\pi/f_\pi} \right) = \\ &= \frac{1}{2} \left[(\partial_\mu h)e^{-i\pi/f_\pi} + (h+v)\frac{(-i)}{f_\pi}(\partial_\mu \pi)e^{-i\pi/f_\pi} \right] \times \\ &\quad \left[(\partial^\mu h)e^{i\pi/f_\pi} + (h+v)\frac{i}{f_\pi}(\partial^\mu \pi)e^{i\pi/f_\pi} \right] = \\ &= \frac{1}{2} \left[(\partial_\mu h)(\partial^\mu h) + \cancel{(h+v)\frac{i}{f_\pi}(\partial_\mu h)(\partial^\mu \pi)} + \cancel{(h+v)\frac{(-i)}{f_\pi}(\partial^\mu h)(\partial_\mu \pi)} \right. \\ &\quad \left. + \frac{(h+v)^2}{f_\pi^2}(\partial_\mu \pi)(\partial^\mu \pi) \right] = \\ &= \frac{1}{2}(\partial_\mu h)(\partial^\mu h) + \frac{(h^2 + v^2 + 2hv)}{2f_\pi^2}(\partial_\mu \pi)(\partial^\mu \pi) = \\ &= \underbrace{\frac{1}{2}(\partial_\mu h)(\partial^\mu h) + \frac{v^2}{2f_\pi^2}(\partial_\mu \pi)(\partial^\mu \pi)}_{\text{pure kinetic terms}} + \underbrace{\frac{(h^2 + 2hv)}{2f_\pi^2}(\partial_\mu \pi)(\partial^\mu \pi)}_{\text{interactions}} \\ &\quad \text{(two fields, two derivatives)} \end{aligned}$$

$\Leftrightarrow$ If we set $f_\pi \equiv v$, we have a canonically normalized kinetic term for $\pi(x)$.

We arrive at the “broken phase” Lagrangian

$$\mathcal{L} = \underbrace{\frac{1}{2}(\partial_\mu h)(\partial^\mu h) - \frac{1}{2}(2\mu^2)h^2 + \frac{1}{2}(\partial_\mu \pi)(\partial^\mu \pi)}_{\text{free Lagrangian}} + \underbrace{\frac{1}{2v^2}(h^2 + 2hv)(\partial_\mu \pi)(\partial^\mu \pi) - \lambda v h^3 - \frac{\lambda}{4}h^4}_{\text{interactions}}$$

As before, we find a massive scalar field, $h(x)$, with mass $m_h^2 = 2\mu^2$, and a massless scalar field, $\pi(x)$.

↔ However, we now interpret the result in terms of symmetry.

Added explanation (not in the original notes)

Why the cross terms cancel. The two struck-out terms are complex conjugates of each other with opposite signs of i : $+\frac{i}{f_\pi}(h+v)(\partial_\mu h)(\partial^\mu \pi)$ and $-\frac{i}{f_\pi}(h+v)(\partial^\mu h)(\partial_\mu \pi)$. Since the two Lorentz contractions are identical, they cancel exactly. This is not an accident either: $\mathcal{L}$ is real, and a term $i \times (\text{real})$ could not survive. Consequently *there is no h - π kinetic mixing*, and the two fields are already the physical, diagonal ones.

Why $f_\pi = v$. The pure kinetic term for π came out as $\frac{v^2}{2f_\pi^2}(\partial\pi)^2$. Canonical normalization means coefficient $\frac{1}{2}$, hence $f_\pi = v$. This is a choice of units for the field π , not new physics: had we kept $f_\pi \neq v$ we would simply rescale $\pi \rightarrow (f_\pi/v)\pi$ at the end. From [p. 19*****] onwards the notes use f_π and v interchangeably — on [p. 21*****] the exponent is written $e^{i\pi/v}$ directly.

Compare the two broken-phase Lagrangians. Part III (Cartesian) and Part IV (exponential) describe the same theory in different variables:

Cartesian: χ, θ	Exponential: h, π
$m_\chi^2 = 2\mu^2, \quad m_\theta = 0$	$m_h^2 = 2\mu^2, \quad m_\pi = 0$
θ appears without derivatives	π appears only via $\partial_\mu \pi$
interactions in V	π interactions in the kinetic term

Field redefinitions do not change S -matrix elements (the “equivalence theorem”), so both give the same physics. The exponential variables are preferred because they make the protecting symmetry manifest.

The derivative coupling has real consequences. Because every π carries a ∂_μ , every vertex carries a factor of momentum, so all amplitudes involving π vanish as its momentum goes to zero. Goldstone bosons decouple at low energy. This “soft theorem” is the same structure met in Lecture 16 for soft photons and Lecture 17 for soft gravitons, and it is the reason the pions of QCD interact weakly at low energy.

19 The shift symmetry [p. 20*****]

The Lagrangian we wrote before, in fact, admits a symmetry. Consider the transformation

$$\begin{aligned} h(x) &\longrightarrow h(x) \\ \pi(x) &\longrightarrow \pi(x) + \text{const} \end{aligned}$$

This is a symmetry since the Lagrangian is invariant.

The key aspect is that $\pi(x)$ only enters in $\mathcal{L}$ via derivatives $\rightsquigarrow$ the constant shift leaves the

Lagrangian invariant.

We call the above symmetry a “shift symmetry”.

OBS. The transformation $\pi(x) \rightarrow \pi(x) + \text{const}$ is non-linear.

$\hookrightarrow$ A linear transformation looks like

$$\pi(x) \rightarrow A\pi(x) = \pi(x) + \alpha\pi(x) + \dots$$

so that the change $\delta\pi = \alpha\pi(x)$ is proportional to the field itself. In the case $\pi(x) \rightarrow \pi(x) + c$, on the contrary, $\delta\pi = \text{const}$ is field independent.

What is the origin of this symmetry? Let us go back to our definition.

Added explanation (not in the original notes)

Verify the invariance directly. In the broken-phase Lagrangian of [p. 19*****] every occurrence of π is inside a $\partial_\mu\pi$: the free term $\frac{1}{2}(\partial\pi)^2$ and the interaction $\frac{1}{2v^2}(h^2 + 2hv)(\partial\pi)^2$. Since $\partial_\mu c = 0$, the shift changes nothing at all — not just up to a total derivative, but identically.

Why “non-linear” is the important word. In a linear realization the fields sit in a representation of the group and the vacuum ($\varphi = 0$) is a fixed point of the action. A shift $\pi \rightarrow \pi + c$ has *no* fixed point: it moves every configuration, including the vacuum $\pi = 0$. That is the algebraic fingerprint of a spontaneously broken symmetry. Compare:

$$\begin{aligned} \text{unbroken (Wigner-Weyl): } \delta\varphi &\propto \varphi \quad (= 0 \text{ at the vacuum}), \\ \text{broken (Nambu-Goldstone): } \delta\pi &= \text{const} \neq 0 \text{ even at } \pi = 0. \end{aligned}$$

This is what protects the mass. A term $m_\pi^2\pi^2$ is *not* invariant under $\pi \rightarrow \pi + c$, so the shift symmetry forbids it — to all orders in perturbation theory, not just at tree level. This answers the question raised on [p. 14*****]: the massless field is massless *because* of a symmetry, and it stays massless in the quantum theory. Fields protected this way are called pseudo-Goldstone bosons when the symmetry is only approximate; the pion mass in QCD is small, not zero, because the chiral symmetry is explicitly broken a little by the quark masses.

Its Noether current. Applying Noether’s theorem (Lecture 13) to $\delta\pi = c$ gives $j^\mu = \partial^\mu\pi \times [1 + (h^2 + 2hv)/v^2]$, whose leading piece is simply $\partial^\mu\pi$. The statement $\langle 0 | j^\mu | \pi(p) \rangle \propto p^\mu \neq 0$ — the current creates the Goldstone boson out of the vacuum — is the technical version of “spontaneously broken”.

20 The origin of the shift symmetry: a non-linear realization of $U(1)$ [p. 21*****]

We write

$$\varrho(x) = \frac{1}{\sqrt{2}} [v + h(x)] e^{i\pi(x)/v}$$

and take the transformation $\begin{cases} h \rightarrow h \\ \pi \rightarrow \pi + c \end{cases}$

$$\begin{aligned} \varrho(x) = \frac{1}{\sqrt{2}} [v + h(x)] e^{i\pi(x)/v} &\longrightarrow \frac{1}{\sqrt{2}} [v + h(x)] e^{i(\pi+c)/v} = \\ &= \underbrace{\frac{1}{\sqrt{2}} [v + h(x)] e^{i\pi/v}}_{= \varrho(x)} e^{ic/v} = e^{ic/v} \varrho(x) \end{aligned}$$

↔ This is nothing but a $U(1)$ transformation on $\varrho(x)$!

We thus find that

$$\begin{aligned} h &\rightarrow h \\ \pi &\rightarrow \pi + \text{const} \end{aligned}$$

is nothing but our original $SO(2) \cong U(1)$ symmetry but now realized non-linearly as a shift symmetry on the field $\pi(x)$.

“non-linear realization of the symmetry”

Added explanation (not in the original notes)

The one-line summary of the whole lecture. The $U(1)$ acts on ϱ by a phase. In polar variables the phase *is* the field π/v . So the symmetry acts by translating π — and a translation of a coordinate along the flat bottom of the potential can obviously cost no energy. Masslessness and symmetry are the same statement seen from two sides.

Why this is called a “realization” and not a “breaking”. The Lagrangian is *exactly* as symmetric after the shift as before; nothing has been broken in $\mathcal{L}$. What changed is how the symmetry is written on the fields: linearly on (ϕ_1, ϕ_2) , as a shift on π . The term “spontaneous symmetry breaking” refers strictly to the *vacuum*, not the Lagrangian — which is exactly what [p. 22*****] stresses.

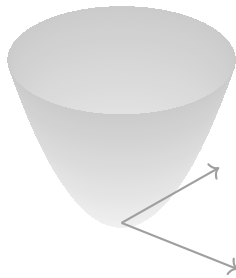
The general theorem behind this instance (Callan, Coleman, Wess, Zumino, 1969): when G breaks to H , the Goldstone fields are coordinates on the coset G/H , and the unbroken H acts linearly on them while the broken generators act as shifts plus field-dependent corrections. Here H is trivial and $G/H = U(1)$, so the shift is exact with no corrections — the simplest possible case, which is why the lecture uses it.

The periodicity, for completeness. π and $\pi + 2\pi v$ give the same ϱ , so strictly π is an angle, and the shift symmetry is $U(1)$ rather than $\mathbb{R}$. Perturbation theory around $\pi = 0$ never sees the difference; the distinction matters only for non-perturbative objects (vortices, winding configurations), which are outside this course.

21 Wigner–Weyl versus spontaneous symmetry breaking: the summary [p. 22*****]

At the conceptual level, it is important to stress that the symmetry $SO(2) \cong U(1)$ is still present in the Lagrangian after spontaneous symmetry breaking (in the “broken phase”), it is only realized in a different, non-linear way.

Wigner–Weyl

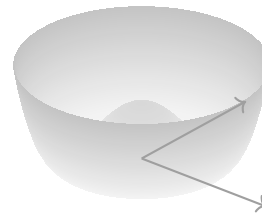


- Symmetry realized linearly on the dynamical fields of the theory

$$\begin{pmatrix} \phi_1 \\ \phi_2 \end{pmatrix} \rightarrow R(\theta) \begin{pmatrix} \phi_1 \\ \phi_2 \end{pmatrix}$$

- Unique vacuum state invariant under the symmetry
- Symmetry realized in the spectrum of the theory

Spontaneous sym. breaking



- Degeneracy of vacua
- The ground state of the theory is not invariant under the symmetry
- Symmetry realized non-linearly in the broken phase
- Presence of a massless scalar field

Added explanation (not in the original notes)

The single most important line on this sheet is the opening one: the symmetry is *not gone*. Everything in $\mathcal{L}$ is still $U(1)$ invariant; the relations it imposes between couplings still hold. What fails is $U(g) |0\rangle = |0\rangle$. This is why the name “breaking” is half-misleading, and why some authors prefer “hidden symmetry”.

What survives, and what does not.

- *Survives*: the Noether current and its conservation $\partial_\mu j^\mu = 0$; the relations among couplings dictated by the symmetry (in the broken-phase Lagrangian of [p. 19*****] the coefficients of h^3 , h^4 and $h(\partial\pi)^2$ are all fixed by the two parameters μ, λ — they are not independent); renormalizability.
- *Does not survive*: the degeneracy of the mass spectrum; the invariance of the vacuum; the possibility of assembling the particles into multiplets of G .

A caveat about the conserved charge. The current is conserved, but the charge $Q = \int d^3x j^0$ does not exist as an operator on the broken vacuum — $Q |0\rangle$ has infinite norm, which is precisely the infinite-volume statement that $Q |0\rangle \neq 0$ would otherwise assert. The interested reader should treat “ $Q |0\rangle \neq 0$ ” as shorthand for “ $\langle 0 | j^\mu | \pi(p) \rangle \neq 0$ ”, which is well defined.

Summary of Lecture 24 — the logic in ten lines

Added explanation (not in the original notes) — Summary

1. Two real scalars packed into $\phi = (\phi_1, \phi_2)^T$ rotating under a *global* $SO(2)$. The only invariant is $\phi^T \phi$, so the most general renormalizable Lagrangian is $\frac{1}{2}(\partial\phi)^T(\partial\phi) - \frac{m^2}{2}\phi^T\phi - \frac{\lambda}{4}(\phi^T\phi)^2$. [p. 1*****]–[p. 2*****]
2. The vacuum is found by minimizing V over constant configurations. Since V depends only on $X = |\phi|$, this is a one-variable problem. [p. 3*****]
3. Of the four sign choices, $\lambda < 0$ is excluded (unbounded below). Two survive: $m^2 > 0$ (unique minimum at the origin) and $m^2 = -\mu^2 < 0$ (minimum on a circle of radius $v = \sqrt{\mu^2/\lambda}$). [p. 4*****]–[p. 6*****]
4. *Case 1* is Wigner–Weyl: the vacuum is invariant, the two scalars are degenerate in mass. Mass degeneracy is the fingerprint of a linearly realized, unbroken symmetry. [p. 6*****]–[p. 7*****]
5. *Case 4* is the Mexican hat: the vacuum manifold is a circle S^1 , a continuum of degenerate ground states related by $SO(2)$. The system must pick one; once picked, the ground state is no longer rotationally invariant. [p. 7*****]–[p. 8*****]
6. Choose $\langle\phi\rangle = (v, 0)^T$ and shift: $\phi_1 = v + \chi$, $\phi_2 = \theta$, so that the new fields have zero vev — necessary for the Fock-space construction of the quantum theory. [p. 9*****]–[p. 11*****]
7. Expanding the potential, the constant and linear terms cancel (by the choice of shift and because we expand about a stationary point), and the quadratic part reads $\frac{1}{2}(2\mu^2)\chi^2 + 0 \cdot \theta^2$: **one massive field** ($m_\chi^2 = 2\mu^2$) **and one massless field**. [p. 12*****]–[p. 13*****]
8. Geometrically: the radial direction climbs the wall (massive); the angular direction runs along the trough of degenerate minima (massless). This is Goldstone’s theorem — one massless scalar per broken generator, here $\dim G - \dim H = 1$. [p. 13*****]
9. Rewriting the theory in polar field variables, $\varrho = \frac{1}{\sqrt{2}}(v+h)e^{i\pi/v}$, makes the reason manifest: V depends only on $\varrho^*\varrho = \frac{1}{2}(v+h)^2$, so π disappears from the potential entirely and appears only through $\partial_\mu\pi$. Canonical normalization fixes $f_\pi = v$. [p. 15*****]–[p. 19*****]
10. The residual symmetry is the shift $\pi \rightarrow \pi + c$ (with $h \rightarrow h$) — which is just the original $U(1)$ phase rotation of ϱ , now realized *non-linearly*. It forbids $m_\pi^2\pi^2$ to all orders. The symmetry is not destroyed by spontaneous breaking; only the *vacuum* fails to respect it. [p. 20*****]–[p. 22*****]

Added explanation (not in the original notes) — Where this is going

This lecture used a *global* symmetry throughout, and its punchline was the appearance of a massless scalar. Lecture 23 ended with the opposite problem: a gauge boson needs a mass, but writing one by hand destroys the theory, and the culprit was a rogue longitudinal polarization with no dynamics to control it.

Put the two together. Repeat this entire construction with the $U(1)$ made *local*, so that ϱ is coupled to a gauge field A_μ through $D_\mu = \partial_\mu + ieA_\mu$. The angular field π can now be removed by a gauge transformation — it is precisely the parameter of the gauge symmetry — and when it disappears from the scalar sector it reappears as the longitudinal polarization

of A_μ , which thereby becomes massive. The counting works out exactly: 2 scalar + 2 photon polarizations = 1 scalar + 3 massive-vector polarizations. The Goldstone boson is “eaten”; the surviving radial field h is the Higgs boson.

That is the Higgs mechanism, and the massless mode derived here is the ingredient it consumes. Everything on these 22 sheets is therefore preparation — which is why the lecture is so insistent that the symmetry is still there, only hidden. A symmetry that had really been destroyed would take renormalizability down with it; a symmetry that is merely non-linearly realized does not.